

Exam 2 Review test.

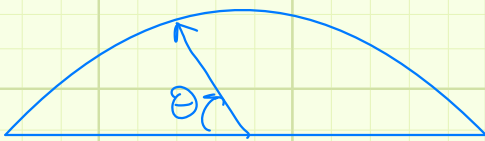
Q.1 (Chp. 6) Thin airfoil Theory

$$\gamma(\theta) = 2\alpha V_\infty \frac{1 + \cos\theta}{\sin\theta}$$

a) show that γ satisfies Kutta condition ($\gamma(\pi) = 0$)

b) What is the meaning of $\frac{2\gamma}{V_\infty}$?

a)



$$0 < \theta < \pi \quad \frac{x}{c} = \frac{1 - \cos\theta}{2}$$

$$\gamma(\pi) = 2\alpha V_\infty \frac{1 + \overbrace{\cos\pi}^{-1}}{\underbrace{\sin\pi}_0} = \frac{0}{0}$$

$\therefore$ L'Hospital

$$\lim_{\theta \rightarrow \pi} \frac{2\alpha V_\infty (1 + \cos\theta)}{\sin\theta} = \lim_{\theta \rightarrow \pi} \frac{2\alpha V_\infty (-\sin\theta)}{\cos\theta} = 0$$

$\therefore$ Kutta condition satisfied.

b) $\gamma(\theta) = 2\alpha V_\infty \frac{1 + \cos\theta}{\sin\theta}$

$$L = \int \rho_\infty V_\infty \gamma(\theta) d\theta$$

$$dL = \rho_\infty V_\infty \gamma(\theta) = 2\alpha \rho_\infty V_\infty^2 \frac{1 + \cos\theta}{\sin\theta}$$

$$L = \int (P_l - P_u) dx \Rightarrow \underbrace{dL}_{\text{Lift per length}} = dp \Rightarrow \frac{dL}{\underbrace{\frac{1}{2} \rho_\infty V_\infty^2}_{q_\infty}} = \frac{2\alpha \rho_\infty V_\infty^2 \left(\frac{1 + \cos\theta}{\sin\theta} \right)}{\frac{1}{2} \rho_\infty V_\infty^2}$$

$$\frac{dl}{q_\infty} = 4\alpha \frac{1+\cos\theta}{\sin\theta} = \frac{dP}{q_\infty} = \frac{P_l - P_u}{q_\infty} = \frac{(P_l - P_\infty) - (P_u - P_\infty)}{q_\infty}$$

$$C_{P_l} - C_{P_u} \Rightarrow C_{P_l} - C_{P_u} = 4\alpha \frac{1+\cos\theta}{\sin\theta} = dl$$

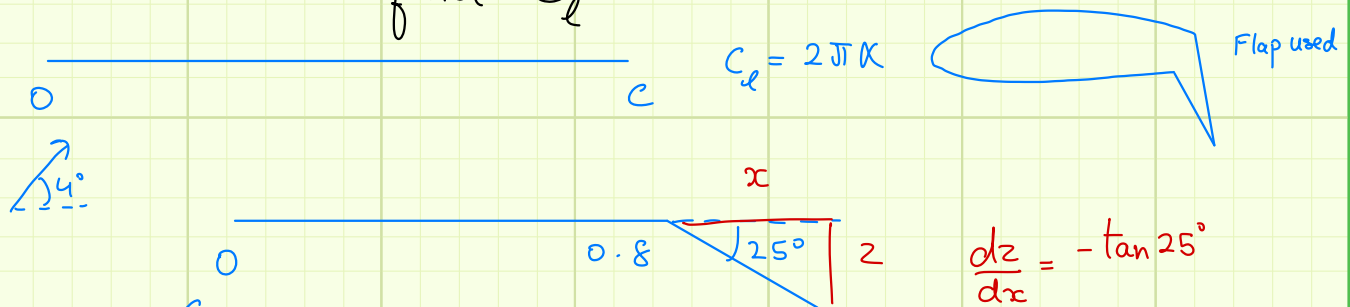
$$\underbrace{\frac{2\gamma(\theta)}{V_\infty}} \quad \underbrace{\frac{2\gamma(\theta)}{V_\infty}}$$

Use the definition of $\gamma(\theta)$

Q2 (Chap - 6)

NACA 0009 with a flap with length $0.2c$ deflected 25°

$\alpha = 4^\circ \rightarrow$ find C_L



$$C_L = 2\pi \left(A_0 + \frac{A_1}{2} \right)$$

$$\therefore A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dz}{dx} d\theta, \quad A_1 = \frac{2}{\pi} \int_0^\pi \frac{dz}{dx} \cos\theta d\theta$$

$$z = \begin{cases} 0 < x < 0.8c, & 0 \\ 0.8c < x < c, & -\tan 25^\circ \\ & = -0.466 \end{cases}$$

$$\frac{x}{c} = \frac{1 - \cos\theta}{2}$$

$$1 - \frac{2x}{c} = \cos\theta$$

$$\theta = \cos^{-1} \left(1 - \frac{2x}{c} \right)$$

$$\theta = \cos^{-1} (1 - 2 \times 0.8) \\ = 2.21 \text{ rad.}$$

$$A_0 = \alpha - \frac{1}{\pi} \int_{2.21}^\pi (-0.466) d\theta \\ = \left(4 \times \frac{\pi}{180} \right) - \frac{1}{\pi} \int_{2.21}^\pi (-0.466) d\theta = 0.2079$$

$$A_1 = \frac{2}{\pi} \int_{2.21}^\pi (-0.466) \cos\theta d\theta = 0.238$$

$$\text{with flaps} \left\{ C_L = 2\pi \left(A_0 + \frac{A_1}{2} \right) = \boxed{2.05} \right.$$

$$\text{No flaps} \left\{ C_L = 2\pi \left(\frac{4\pi}{180} \right) = 0.438 \right.$$

Chap. 7

7.1 An Airplane,

load distribution: elliptic

$$W = 14700 \text{ N}$$

$$U_{\text{cruise}} = 300 \text{ km/h}$$

$$\Gamma_0 = ?$$

For elliptic wing

$$h = 3000 \text{ m} \rightarrow \rho_{\infty} = 0.9 \text{ Kg}$$

$$S = 17 \text{ m}^2$$

$$AR = 6.2$$

$$\alpha_{L_0} = -1^\circ$$

$$U_{\infty} \left(\frac{\text{m}}{\text{s}} \right) = 83.3 \frac{\text{m}}{\text{s}}$$

$$L = \frac{\pi}{4} b \rho_{\infty} U_{\infty} \Gamma_0$$

$$L = \rho_{\infty} U_{\infty} \Gamma_0 e \frac{\pi}{2} S$$

$$q_{\infty} \left(\text{dynamic pressure} \right) = \frac{1}{2} \rho_{\infty} U_{\infty}^2 = 3122.5 \frac{\text{N}}{\text{m}^2}$$

$$AR = \frac{b^2}{S} \Rightarrow b = \sqrt{6.2 \times 17} = 10.266 \text{ m}$$

$$C_L = \frac{L = W}{\frac{1}{2} \rho U_{\infty}^2 S} = \frac{\pi b \Gamma_0}{2 U_{\infty} S}$$

$$\Gamma_0 = 24.316$$

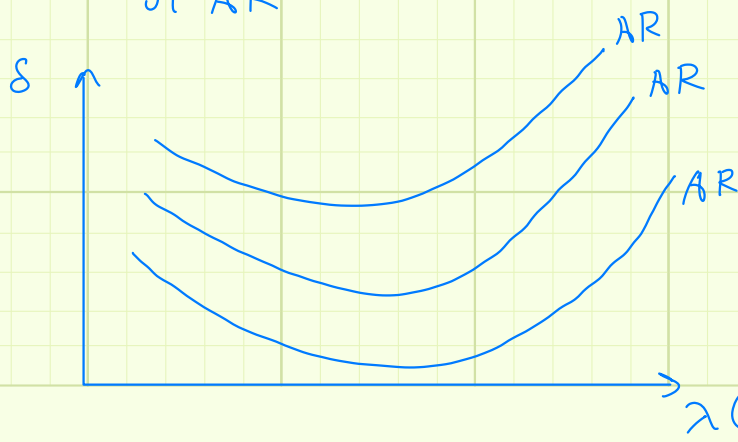
$$\therefore \Gamma_0 = \frac{C_L \times 2 U_{\infty} S}{\pi b}$$

$$C_L = 0.2769$$

$$\Gamma_0 = 24.31620$$

$$C_L = \frac{L}{\rho_{\infty} S} = \frac{W}{\rho_{\infty} S} = 0.2769$$

$$C_{D_v} = \frac{C_L^2}{\pi AR} = 0.003936$$



$$\epsilon = \frac{C_L}{\pi AR} = 0.8057^\circ$$

$$C_L = 2\pi (\alpha_{eff} - \alpha_{L_0}) = 0.2739$$

Airfoil sees the $\alpha_{effective}$

$$\begin{cases} \alpha_{eff.} - \alpha_{L_0} = 2.4977^\circ \\ = 0.04359 \text{ rad} \end{cases}$$

$$\alpha_{eff} = \alpha_{L_0} + 2.4977^\circ$$

$$= -1.2 + 2.4977$$

$$\boxed{\alpha_{eff} = 1.2977}$$

$$\alpha_{eff} = \alpha - \epsilon$$

$$\alpha = \alpha_{eff} + \epsilon = 1.2977 + 0.8057 = 2.1034$$

$$7.2 \quad \Gamma(y) = \Gamma_0 \left(1 - \frac{y^2}{s^2}\right)$$

Assume the above Γ gives the same lift as elliptic wing, obtain Γ_0

$$\begin{aligned} L &= \int_{-s}^s \rho_{\infty} U_{\infty} \Gamma(y) dy = \rho_{\infty} U_{\infty} \Gamma_0 \int_{-s}^s \left(1 - \frac{y^2}{s^2}\right) dy \\ &= \rho_{\infty} U_{\infty} \Gamma_0 \left[y - \frac{y^3}{3s^2} \right]_{-s}^s \\ &= \rho_{\infty} U_{\infty} \Gamma_0 \frac{4s}{3} \end{aligned}$$

$$\rho_{\infty} U_{\infty} \Gamma_0 \frac{4s}{3} = \rho_{\infty} U_{\infty} \Gamma_e \frac{\pi}{2} s$$

$$\Gamma_0 = \frac{3\pi}{8} \Gamma_e$$

$$w_{y_1} = \frac{1}{4\pi} \int_{-s}^s \frac{\frac{d\Gamma}{dy}}{y - y_1} dy$$

Downwash relation.

obtain downwash at the plane of symmetry

